

# V&V3 — Conceptual Validity

## Simulation lifecycle + distribution justification

Figure 2: Order lifecycle (SimPy processes)



### Distribution + resource justification

- Driver: Arena-style fitted from 167,784 ZWM92 dispatch rows (40,804 kit-orders).
- IAT ~ Exp(5.36 min within-shift) — within-shift filter excludes overnight gaps.
- n\_items ~ Empirical(distinct picks per kit, mean 4.0) — clipped at 50.
- Line ~ Categorical (9 product families); material ~ Categorical|line.
- Pick times ~ Lognormal(F400 mean,  $\sigma$  from F400 CV): operator 0.113, RT 0.110,
- manual 0.102, Kardex 0.113 min.  $\sigma$  values: 1.30/1.20/1.40/1.30 (log-scale).
- Resources: 7 RT + 8 op + 7 milk-run + 4 Kardex (verbal report May 2026).
- Position lock (1-cap SimPy resource per pallet position) prevents race conditions.